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RAIPUR (C.G.)

C000619(022)

B. Tech. (Sixth Semester) Examination, Nov.-Dec. 2024

(AICTE Scheme)

(CSE Engg. Branch)

CRYPTOGRAPHY and NETWORK SECURITY

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : *Attempt all question. Part (a) is compulsory and from part (b), (c) & (d) attempt any two questions. Part (a) carry 4 marks and part (b), (c) & (d) each one carries 8 marks.*

Unit-I

1. (a) What is the role of Cryptography?

(b) Differentiate between Substitution Technique and the Transposition Technique.

(c) Explain the types of Security attacks in detail.

(d) Explain the Block Cipher Design Principles?

Unit-II

2. (a) What do you understand by stream cipher?

(b) Discuss Modular arithmetic operation and properties with an example.

(c) Describe the working of RC4 with its advantages and disadvantages.

(d) Identify the steps involved in Euclidian algorithm with suitable example.

Unit-III

3. (a) State Fermat's and Euler's Theorems.

(b) Identify the steps involved in RSA algorithm with suitable example.

(c) What is Key management? What are the different types of public key distribution technique?

(d) Discuss the Diffie-Hellman Key Exchange algorithm in detail.

Unit-IV

4. (a) Define Hash function.

(b) Explain Authentication protocols in detail.

(c) Discuss various approaches of producing authentication method in detail.

(d) How to use public key infra-structure to provide digital signature?

Unit-V

5. (a) What is firewall?

(b) Discuss firewall design principles in detail.

(c) Describe SSL and TSL architecture with suitable diagram.

(d) How you can achieve Electronic mail security?